Introduction to Artificial Intelligence and Machine Learning (CSN11601)

Assignment-4

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Q1. Write a program to compute LCM of a number.
A1. CODE:

```
num1=int(input("Enter first number: "))
num2=int(input("Enter second number: "))
if num1>num2:
   greater=num1
else:
   greater=num2
while True:
   if greater%num1==0 and greater%num2==0:
      lcm=greater
      break
   greater+=1
print("LCM of",num1,"and",num2,"is",lcm)
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p1.py
Enter first number: 12
Enter second number: 34
LCM of 12 and 34 is 204
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p1.py
Enter first number: 20
Enter second number: 5
LCM of 20 and 5 is 20
```

Q2. Write a program to create a multiplication Table.
A2. CODE:

```
n = int(input("Enter a number: "))
for i in range(1,11):
   print(f"{n} x {i} = {n*i}")
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p2.py
Enter a number: 13
13 x 1 = 13
13 x 2 = 26
13 x 3 = 39
13 x 4 = 52
13 x 5 = 65
13 x 6 = 78
13 x 7 = 91
13 x 8 = 104
13 x 9 = 117
13 x 10 = 130
```

Q3. Write a program to print a given string is a palindrome or not.
A3. CODE:

```
s=input("Enter a string: ")
if s == s[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p3.py
Enter a string: madam
Palindrome
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p3.py
Enter a string: artificial
Not Palindrome
```

Q4. Write a program to remove punctuations from a string.
A4. CODE:

```
s = input("Enter a string: ")

punctuations = '`~!@#$%^&*()-_=+[]{};:<>,.?/'
result = ""
for ch in s:
    if ch not in punctuations:
        result+=ch

print("String without punctuations:",result)
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p4.py
Enter a string: Aayansh
String without punctuations: Aayansh
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p4.py
Enter a string: Aayansh i-s so..,,lving>>>< Assign()ment
String without punctuations: Aayansh is solving Assignment
```

Q5. Write a program to print number of vowels present in a string.
A5. CODE:

```
s=input("Enter a string: ").lower()
vowels="aeiou"
count=sum(1 for ch in s if ch in vowels)
print("Number of vowels:",count)
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/AI-ML/Assignment-4$ python3 p5.py
Enter a string: Artifical Intelligence
Number of vowels: 9
```

Q6. Write a Python program to compute the number of characters, words and lines in a file.

A6. CODE:

```
f_name=input("Enter filename: ")
f=open(f_name,"r")
text=f.read()
f.seek(0)
lines=f.readlines()
words=text.split()
cha=len(text)

print("Characters:",cha)
print("Words:",len(words))
print("Lines:",len(lines))
```

OUTPUT:

